13.5 Asking Awkward Questions

WORKSHEET

The question

STEP II 2023, Question 11

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Y is X_1 with probability p and X_2 with probability q. What is $\mathrm{E}(Y)$? What is $\mathrm{E}(Y^2)$?

STAGE 2 Independent randomisation

Ask yourself What is $\mathrm{P}(B = 1)$? Are the n values of B independent?

STAGE 3 One coin for everybody

Ask yourself Now one coin toss decides for all n dice. Given heads, what is the distribution of Z_2? Given tails? Use part (i).

Ask yourself Where is most of the probability? Is there much near the mean $\frac{n}{2}$?

Try this next

1 **The twins.** (From STEP II, 2005.) Twins share a computer; the one online is Anna with probability p, who answers truthfully with probability a, or Bella, who does so with probability b. I ask them to toss a coin and say whether it came down heads, and get "yes". Show that the probability that it was heads is $\frac{1}{2}$ if and only if $2(ap + bq) = 1$. If I then ask again and get "no", show that the probability of heads is $\frac{1}{2}$; if I get "yes" again, show that it is $\frac{1}{2}$ provided $2(ap + bq) = 1$.

2 **Estimating π.** In a Warner survey with $p = \frac{3}{4}$, 400 people answer and 160 say yes. Estimate the proportion π in group A, and find the approximate standard error of your estimate.

3 **Total variance.** Prove the law of total variance for a discrete hidden variable C with several values.

4 **Shared coins.** In part (iii), what is the probability that $Z_2 > \frac{n}{2}$? Explain why, for large n, it is close to $\frac{1}{2}$ even though Z_2 is almost never close to $\frac{n}{2}$.

Further reading

► **Stanley Warner, "Randomized response: a survey technique for eliminating evasive answer bias"**, *Journal of the American Statistical Association* 60 (1965).

- **Cynthia Dwork & Aaron Roth, *The Algorithmic Foundations of Differential Privacy*** (free online). The modern theory; the first chapters are accessible.
- **David Spiegelhalter, *The Art of Statistics*** (Pelican), on surveys and their pitfalls.

The full guided solution, and what the question leads to at university, is in Section 13.5:
https://beyondstep.keithmcnulty.org/ch13/awkward-questions.html